NIAGARA 4 SUPERVISOR & JACE | OFFICIAL STEP-BY-STEP USER GUIDE

TecumSMS Notification Service & Driver (TecumSMS-rt)

SMS Subscription Website API Keys, Niagara Email-to-SMS, Open-Source Gateways, Point Rules & Alarms

Module: TecumSMS-rt.jar **Vendor: Tecum** **Version: 1.0.0** **Palette Symbol: tsms** **Target: Niagara 4.0 - 4.14+**

1. Architecture & Component Overview

The TecumSMS Notification Service & Driver (TecumSMS-rt) adds native internet SMS subscription APIs, Niagara Email-to-SMS, and network SMS alerting to Niagara 4 Supervisor and JACE stations without requiring any serial port modem hardware. Drop SmsService directly under Station -> Config -> Services (recommended), or drop SmsNetwork under Station -> Config -> Drivers.

```
Station (station:|slot:/)
 +-- Services
 |    +-- SmsService (tsms:SmsService - or SmsNetwork under Drivers)
 |    |    +-- [Built-in Subscription Website API Key & Niagara Email-to-SMS options]
 |    |    +-- TextBeeCloud      (tsms:HttpGateway - Open-Source Internet SMS)
 |    |    +-- EmailSmsGateway   (tsms:EmailSmsGateway - Niagara EmailService / SMTP SMS)
 |    |    +-- Contacts          (tsms:SmsContactFolder - Phone numbers & On-Duty hours)
 |    |    +-- Groups            (tsms:SmsGroupFolder - Named teams e.g. Maintenance)
 |    |    +-- Rules             (tsms:SmsRuleFolder - Point rules & Wire-sheet senders)
 |    +-- EmailService           (Niagara EmailService used when Email-to-SMS is enabled)
 |    +-- AlarmService
 |         +-- SmsAlarmRecipient (tsms:SmsAlarmRecipient - Niagara AlarmClass router)
```

2. Step 1 - Install TecumSMS-rt.jar & Trust the Certificate

Step 1.1: Install TecumSMS-rt.jar into Niagara 4

1. Copy dist\TecumSMS-rt.jar from the module package folder into your Niagara installation's modules directory (for example, C:\Niagara\Niagara-4.x.x.x\modules\TecumSMS-rt.jar).
2. Restart Niagara Workbench and your Niagara Station (or Platform Daemon) so the module and palette are loaded.

Step 1.2: Import TecumRootCA.pem into Both Workbench & Platform Trust Stores

1. Workbench Trust Store (My Host): In Niagara Workbench, open My Host -> Certificate Management (or Tools -> Certificate Management), select the User Trust Store tab, click Import, select certs\TecumRootCA.pem (or TecumRootCA.cer) from the module package folder, and click OK.
2. Platform / Station Trust Store (Platform): In the Nav tree, expand Platform -> Certificate Management (or Station -> Config -> Services -> PlatformServices -> CertManagerService), open the User Trust Store tab, click Import, select certs\TecumRootCA.pem, and click OK.
3. Restart your Niagara Station from Platform -> Application Director so the station trusts TecumSMS-rt.jar when adding SmsService.

Step 1.3: Open the TecumSMS Palette in Workbench

1. In Niagara Workbench, open the Palette side bar (Window -> Side Bars -> Palette).
2. Click the Open Palette folder icon, search for TecumSMS, select TecumSMS, and click OK. At the top of the palette you will see SmsService (for Services) and SmsNetwork (for Drivers).

3. Step 2 - Configure SmsService (Subscription API Key & Niagara Email-to-SMS)

Step 2.1: Add SmsService to Station -> Config -> Services

1. In the Nav tree, expand Station (<StationName>) -> Config -> Services.
2. Drag SmsService from the top of the TecumSMS palette and drop it onto Services (or drag SmsNetwork onto Config -> Drivers).
3. Double-click SmsService to open its Property Sheet. You can enter your SMS Subscription Website API Key or enable Niagara Email-to-SMS directly on the SmsService Property Sheet:

SmsService Property	Default	Purpose
`Subscription Preset`	TextBee Cloud	Select your SMS subscription website (TextBee Cloud, SMS-Gate Cloud, Traccar SMS, ntfy.sh, Twilio, ClickSend, Vonage/Nexmo, Textbelt, MessageBird, Jasmin, playSMS, Kannel, Custom).
`Subscription Api Key`	(empty)	API Key or Auth Token for your SMS subscription website. Automatically shared with child gateways or used directly by SmsService.
`Subscription Account Id`	(empty)	Account SID, Device ID, Topic, or Username for your SMS subscription website.
`Subscription Sender Id`	(empty)	Sender ID or From phone number (used by Twilio, ClickSend, Vonage, MessageBird).
`Subscription Api Url`	(empty)	Optional custom endpoint URL (leave blank to use the selected preset's default URL).
`Enable Niagara Email Sms`	false	Set to true to send SMS through Niagara's built-in EmailService (Email-to-SMS).
`Email Service Ord`	station:|slot:/Services/EmailService	ORD of Niagara's EmailService or a specific OutgoingAccount inside EmailService.
`Email Sms Domain`	sms.clicksend.com	Email-to-SMS gateway or carrier domain (e.g. sms.clicksend.com, textmagic.com, vtext.com, txt.att.net).
`Email Sms To Format`	{phoneDigits}@{domain}	Recipient email template ({phoneDigits}@{domain}, {phone}@{domain}, or fixed address).

4. Step 3 - Configure Internet SMS Gateways or EmailSmsGateway

In addition to the built-in SmsService settings, you can expand Palette -> Gateways and drop one or more dedicated gateway components onto SmsService for multi-gateway priority failover.

Option A: Open-Source & Commercial HTTP Subscription Gateways (with API Key)

1. Expand Gateways in the TecumSMS palette and drag your preferred gateway onto SmsService: TextBeeCloud, SmsGateCloud, TraccarSmsCloud, NtfyInternetAlert, TwilioSmsGateway, ClickSendSmsGateway, VonageSmsGateway, TextbeltSmsGateway, or MessageBirdSmsGateway.
2. Open the gateway's Property Sheet: paste your subscription website API Key into Api Key, and enter your Device ID / Account SID / Username into Username (and Sender Id if required).
3. Click Save. Right-click the gateway -> Actions -> Send Test SMS, enter a phone number (+countrycode...), and click OK.

Option B: Send SMS via Niagara Email (EmailSmsGateway or Built-in Option)

1. Ensure Niagara's EmailService is configured under Station -> Config -> Services -> EmailService with a valid OutgoingAccount.
2. Either set Enable Niagara Email Sms = true directly on SmsService, OR drag EmailSmsGateway from Palette -> Gateways onto SmsService.
3. Set Sms Gateway Domain to your Email-to-SMS provider or carrier domain (e.g. sms.clicksend.com, textmagic.com, vtext.com, txt.att.net, tmomail.net) and To Address Format (default {phoneDigits}@{domain}).
4. If your Email-to-SMS provider requires an API key in the email subject or body, enter it in Api Key and reference {apiKey} in Subject Template or Body Template.
5. Note: EmailSmsGateway also supports direct SMTP/SMTPS without Niagara EmailService by setting Use Niagara EmailService = false and entering SMTP Host, Port, Username, and Password.

All Pre-Configured Gateways in Palette -> Gateways

Palette Component	Platform / Website	API Key & Property Sheet Configuration
`EmailSmsGateway`	Niagara EmailService / Direct SMTP	Routes SMS via Niagara EmailService (station:|slot:/Services/EmailService) or direct SMTP to {phoneDigits}@{domain}.
`TextBeeCloud`	textbee.dev (Open-Source MIT)	Set Username = Device ID, Api Key = TextBee API Key.
`SmsGateCloud`	api.sms-gate.app (Open-Source)	Set Username and Api Key (or Password) from the Android SMS Gateway app.
`TraccarSmsCloud`	traccar.org/sms (Open-Source)	Set Api Key = Traccar Cloud API Token.
`NtfyInternetAlert`	ntfy.sh (Zero-SIM Push Alert)	Set Username = ntfy topic name (and optional Api Key = access token).
`TwilioSmsGateway`	Twilio SMS Subscription	Set Username = Account SID, Api Key = Auth Token, Sender Id = Twilio From number.
`ClickSendSmsGateway`	ClickSend SMS Subscription	Set Username = ClickSend Username, Api Key = ClickSend API Key, Sender Id = From.
`VonageSmsGateway`	Vonage / Nexmo Subscription	Set Username = API Key, Api Key = API Secret, Sender Id = From.
`TextbeltSmsGateway`	Textbelt SMS API	Set Api Key = Textbelt API Key.
`MessageBirdSmsGateway`	MessageBird SMS Subscription	Set Api Key = Live Access Key, Sender Id = Originator.
`SmppGateway`	SMPP v3.4 (Jasmin / Kannel / Telecom)	Set Address (host), Port (2775), Use TLS, System ID, Password, and Sender ID.

5. Step 4 - Add Contacts & Groups (Address Book)

Step 4.1: Add Contacts & Groups under SmsService

1. Expand SmsService -> Contacts, drag SmsContact from the TecumSMS palette into Contacts, set Phone Number in international format (e.g. +971501234567), and configure On Duty Days / Time.
2. Expand SmsService -> Groups, drag SmsGroup into Groups, open its Property Sheet, expand Members, and tick the checkboxes for each contact in that group.

6. Step 5 - Configure Point Rules, Wire-Sheet SmsSender & AlarmService

Step 5.1: Point Rules (SmsPointRule), Wire-Sheet (SmsSender) & Alarms (SmsAlarmRecipient)

1. Point Monitoring Rules: Expand SmsService -> Rules and drag SmsPointRule (or HighValueRule, StateRule, OfflineRule) into Rules. Select Source ORD, Condition, Limit / Compare Value, Active Delay, and tick Recipients.
2. Wire-Sheet Triggers: Drag SmsSender from Palette -> Rules onto any Wire Sheet and link a boolean output to Trigger.
3. Niagara Alarm Routing: Open Station -> Config -> Services -> AlarmService (Wire Sheet), drag SmsAlarmRecipient from the TecumSMS palette, link alarmEscalation1 from your AlarmClass to routeAlarm, and tick Recipients.

7. Step 6 - Verification, Diagnostics & Troubleshooting

Open the Property Sheet of SmsService at any time to inspect Queue Size, Total Sent, Total Failed, Last Sent Time, Last Error, and the live 50-line Activity log.

Diagnostic Message	Resolution
`Missing module TecumSMS`	Import certs\TecumRootCA.pem into Platform -> Certificate Management -> User Trust Store (not only My Host), verify TecumSMS-rt.jar is in C:\Niagara\Niagara-4.x.x.x\modules\, and restart the Station.
`no reachable recipients`	Tick at least one Contact or Group under Recipients, verify Phone Number is valid (+countrycode...), and check On Duty Days / Time.
`HTTP 401 / HTTP 403`	Check Api Key and Username (Device ID / Account SID) on SmsService or the HttpGateway Property Sheet.
`Niagara EmailService not found`	Ensure EmailService is added under Station -> Config -> Services -> EmailService with a valid OutgoingAccount, or set Use Niagara EmailService = false on EmailSmsGateway to use direct SMTP.